

HENRY HSU

Houston, TX 77030

+1-510-996-6837

www.yarikama.com

yarikama

yarikama

hsuhengjui@gmail.com

SUMMARY

Product-focused SWE across TypeScript, React, Node.js, PostgreSQL, Elasticsearch. Built integration test infra and internal LLM tooling at Google; as backend eng and team lead, optimized 140+ APIs (27.7% faster), scaled async pipelines 3M→20M+ records, raised coverage 0→67%. 14 merged PRs to LlamaIndex (15K+ stars). Rice M.C.S., Dec 2026.

WORK EXPERIENCE

Software Engineering Intern

Jun. 2026 – Aug. 2026

Google

Taipei, Taiwan

- **Test Infrastructure** – Built 3 integration test infrastructures from a **zero** baseline (**Java, Python**), onboarding **400+** test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Automated Failure Triage** – Built an LLM-backed service that triages scheduled integration test runs and posts root-cause summaries directly onto issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Reproduction Tooling** – Extended the service to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Back-End Engineer & GenAI Team Lead

Sep. 2024 – Aug. 2025

MaiAgent Co., Ltd. – Award-Winning B2B GenAI Startup

Taipei, Taiwan

- **API Optimization** – Optimized **140+** RESTful APIs through SQL query refactoring, connection pooling (**Elasticsearch**, Cohere, OpenAI), and Django caching; cut response time of 13 high-traffic APIs by **27.7%** overall and eliminated N+1 queries
- **Data Pipeline & Scaling** – Architected an async processing pipeline with **asyncio**, **Celery**, and batch/generator patterns for multi-level knowledge base indexing, achieving **5x faster** parsing while scaling from **3M to 20M+** records through a normalized **PostgreSQL** schema and backward-compatible incremental refactors
- **Real-Time Communication** – Implemented a **WebSocket** notification system with **Redis** pub/sub for file parsing status; built event-driven architecture for agent state transitions, pushing render triggers to the frontend for live UI updates
- **CI/CD & Testing** – Built a **GitHub Actions** CI pipeline with pytest unit and E2E coverage, reaching **67%** code coverage from a **zero** baseline; reduced production hotfixes by **90%** initially and **50%** long term
- **Open Source** – Contributed **14** merged pull requests to **LlamaIndex (15K+ stars)**, fixing bugs and adding features across the AWS Bedrock, Claude, **Elasticsearch**, Cohere, OpenAI, and MCP Client integrations

EDUCATION

Rice University - Top 20 U.S. University

Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00)

Houston, TX

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University

Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List)

Hsinchu, Taiwan

Minor: Computer Science (Domain GPA: 4.13/4.30)

PROJECTS

Elderly Care Map Website | Apache, PHP, MySQL, Google Maps API, Bootstrap5, JavaScript

Sep. 2023 – Jan. 2024

- **Database Design & ETL**: Designed a normalized 8-table schema covering 2,000+ long-term care institutions with joins and constraints for user favorites and geolocation data; integrated government open data (CSV) through an automated ETL pipeline
- **Full-Stack Development**: Built a map-based search platform on the Google Maps API with radius filtering, color-coded bed-capacity indicators, and real-time distance calculation; implemented user authentication and CRUD for saved institutions

Chat Bar - Real-Time Multiplayer Game Server | C/C++ (Socket Programming, SFML), MySQL, SHA-256

Oct. 2023 – Jan. 2024

- **Server-Client Architecture**: Designed a real-time multiplayer game server using **TCP socket programming** with multi-threaded client handling, supporting concurrent connections for chat and gameplay state synchronization
- **User Management & Client**: Built authentication with SHA-256 password hashing and MySQL persistence plus ranking features backed by optimized queries; developed the SFML client with an event-driven loop for movement, group chat, and live state updates

AWARDS

Presidential Hackathon Winner (2024) – Urban noise detection with LLM-powered structured analysis

23rd Golden Peak Award (2025) – Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: TypeScript, JavaScript, Python, Java, C/C++ | **Frontend**: React, HTML, CSS | **Backend**: Node.js, Express.js, Django, FastAPI

Databases: PostgreSQL, Elasticsearch, MongoDB, Redis, Neo4j | **APIs**: RESTful, WebSocket, Redis Pub/Sub | **Async**: asyncio, Celery

Testing: pytest, JUnit | **DevOps**: GitHub Actions, Docker, git, Nginx | **AWS**: EC2, S3, RDS | **AI Tools**: Claude Code, Codex